Detecting AI-Generated Text Using Lightweight Linguistic Models

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Introduction

As LLMs like ChatGPT become more capable, it's getting harder to tell AI-generated text apart from human writing. This matters for academic integrity, journalism, and trust online.

We ask: **can a simple, interpretable model do the job?** Instead of a complex neural detector, we use logistic regression with TF-IDF features.

Research Questions:

- Can lightweight models reliably detect AI-generated text?
- Which linguistic patterns most distinguish ChatGPT from humans?
- Does detection hold across different topic domains?

Accuracy by Source

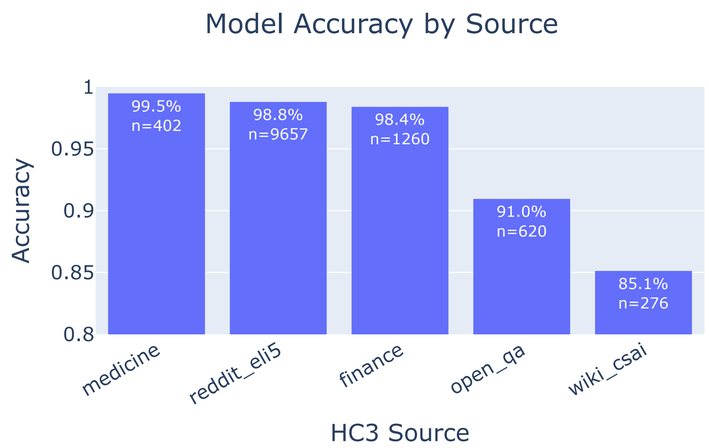

Figure 1. All five HC3 domains exceed 85% accuracy, indicating the model learns **authorship style**, not domain-specific vocabulary.

Key Insight

Simple lexical features are enough to separate human and AI writing with high confidence.

Methodology

Dataset (HC3):

- Human and ChatGPT responses to identical questions
- Each sample is a (question, answer) pair labeled as human or AI
- Binary classification: human (0) vs ChatGPT (1)

Domains:

- Reddit (ELI5) — informal explanations
- Open QA — general knowledge questions
- Wikipedia (CS) — technical explanations
- Medicine — clinical Q&A
- Finance — investment discussion

Model:

- TF-IDF (50k features, unigrams + bigrams)
- Logistic Regression
- Train/val/test split: 75% / 10% / 15%

Model Confidence Distribution

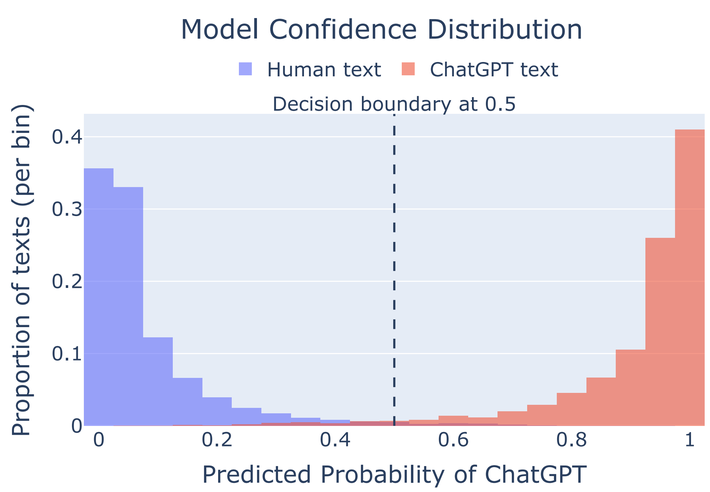

Figure 2. 97.4% of ChatGPT texts are correctly classified; 1.6% of human texts are misclassified. Clear separation at 0.5 shows **high-confidence predictions**.

Feature Importance

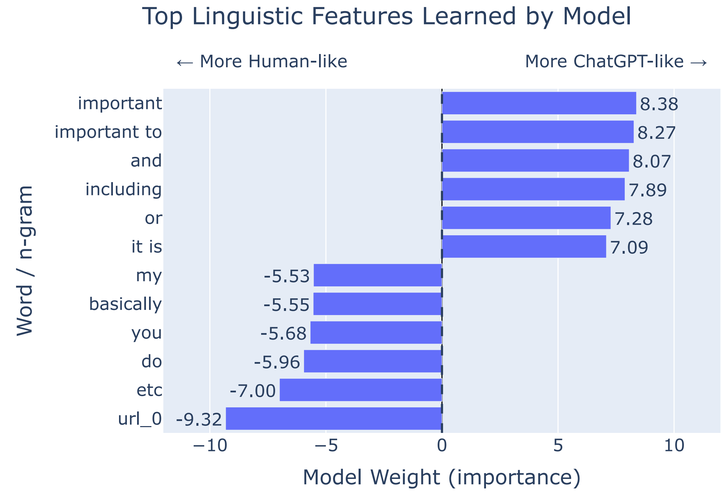

Figure 3. ChatGPT-indicative features (right) include common, generic terms such as *"important," "including,"* and *"it is."* Human-indicative features (left) include more informal and personal language like *"my," "you,"* and *"basically."* This contrast helps explain why a linear model separates the two effectively.

Findings

- >**95% accuracy** on held-out test data
- ChatGPT text is **stylistically uniform**, while human text is more diverse
- ChatGPT features skew toward **common, generic terms**, while human features reflect more personal and informal language
- Performance generalizes across all five domains

Takeaways:

- Lightweight models achieve strong performance at a fraction of the cost of neural detectors
- Interpretability allows us to **explain model decisions**, which is valuable in real-world settings
- As LLMs improve, lexical signals may weaken, requiring more advanced stylistic features

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